

FPGA Hardware vs PyTorch Validation Report (Corrected)

Multi-Head Self-Attention (MHSA) Accelerator — ViT-Base, 12 heads, 197 tokens, 768-dim

1. Objective

Verify that the FPGA MHSA accelerator output matches a PyTorch floating-point reference, using the full ViT-Base workload (197 tokens x 768 dims, 12 heads x 64 dim).

2. Validation Methodology

Step 1 – Reference generation: PyTorch produced q.txt, k.txt, v.txt, golden_output.txt.

Step 2 – Hardware execution: mhsa_top() computes $\text{Softmax}(\mathbf{QK}^T/\sqrt{d})\mathbf{V}$ via AXI, using the online-softmax architecture. Scale register = 512, interpreted in Q4.12 as $512/4096 = 0.125 = 1/\sqrt{64}$ (confirmed correct).

Step 3 – Fixed-point conversion (CORRECTED):

- Inputs: Q4.12 format, scale factor 4096 ($Q_{\text{int16}} = \text{round}(Q_{\text{float}} \times 4096)$) — confirmed correct against notebook cell 60.
- Output: empirically requires dividing the raw int32 by 2^{22} (4,194,304), NOT by 2^{16} (65,536) as originally written. This implies ~22 fractional bits on the output port, not 16 — i.e. closer to ap_fixed<24,2> than the ap_fixed<24,8> assumed in earlier reports.
- This divisor (2^{22}) was reverse-engineered by ratio-matching against the golden reference (see Section 8), not read directly from the HLS ap_fixed<> typedef. Recommended before publishing: confirm the exact output type in the HLS C++ source to rule out a coincidental scale match.

3. Validation Dataset Size

Parameter	Value
Tokens	197
Embedding Dimension	768
Total Outputs	151,296 (197×768)

4. Hardware vs PyTorch Results

Metric	Result
Mean Absolute Error (MAE)	0.00189344
RMSE	0.00978432
Maximum Error	0.824045
Worst-case location	Token 59, Head 4, Dim 43 (flat index 45,611 — verified by re-deriving token/head/dim from the index)

5. Output Match Analysis

Outputs with absolute error > 0.1: 99 of 151,296 → match rate = $(151,296 - 99) / 151,296 \times 100 = 99.9346\%$.

Note: counts for the >0.5 and >1.0 thresholds are computed in the notebook (cell 80) but not captured in this write-up — include them for a fuller picture of the error tail before final submission.

6. Accuracy Relative to PyTorch

Normalized RMSE = $\text{RMSE} / \max|\text{golden output}| = 0.009784 / 3.4081 \times 100 = 0.287\%$. Accuracy = $100 - 0.287 = 99.713\% \approx 99.7\%$.

7. Per-Head Error Analysis

Head	Mean Error
0	0.00378
1	0.00031
2	0.00089

3	0.00027
4	0.01358
5	0.00124
6	0.00104
7	0.00029
8	0.00066
9	0.00033
10	0.00018
11	0.00016

Head 4 has the highest deviation and contains the single worst-case error (Token 59). All other heads are an order of magnitude lower — worth a targeted look at Head 4's data path, but not indicative of a systemic bug across heads.

8. De-quantization Investigation — Corrected Narrative

The original report said dividing by 65,536 resolved the mismatch. Tracing the notebook shows this is inaccurate; the actual debugging sequence was:

- Divide by 65,536 (Q16 assumption) → large residual mismatch remained.
- Ratio analysis: $\text{mean(hw)}/\text{mean(gold)}$ and $\text{max(hw)}/\text{max(gold)}$ both landed near 64, and hw/gold elementwise mean ratio ≈ 63.6 — i.e. dividing by 65,536 alone left the result exactly $\sim 64\times$ too large.
- Trying hw/64 on top of the /65,536 result closed most of the gap — equivalent to dividing the raw output by $65,536 \times 64 = 4,194,304 = 2^{22}$.
- Final corrected conversion: `hw_correct = O.view(int32).astype(float64) / (2**22)`. This is the version used for every metric in Sections 4–7 above.

Conclusion: the mismatch was a de-quantization/scale error, not a functional hardware bug — consistent with the earlier report's conclusion — but the correct fix is a $/2^{22}$ scale, not $/2^{16}$. The methodology and root-cause sections have been corrected accordingly.

9. Cross-Check Against Test Harness (Prior Audit Item)

The earlier audit flagged that the notebook never loaded real `q_np/k_np/v_np` into the hardware buffers before comparison. This is now fixed: cell 60 explicitly quantizes and writes `q_np/k_np/v_np` into Q/K/V before the run in cells 61–66. This resolves that concern — the results in this report are from a real end-to-end run against the actual test vectors, not a stale synthetic buffer.

10. Conclusion

The FPGA MHSA accelerator shows strong agreement with the PyTorch floating-point reference: $\sim 99.7\%$ numerical accuracy, 99.93% of outputs within ± 0.1 , MAE 0.0019, RMSE 0.0098. This is a genuine, harness-verified result. Before treating this as final for publication/faculty sign-off, two follow-ups remain: (1) confirm the output's true fractional-bit width against the HLS `ap_fixed<>` source rather than the empirically reverse-engineered 2^{22} divisor, and (2) investigate why Head 4 / Token 59 shows a localized error $\sim 8\times$ the next-highest value.